WHERE VIOLENCE PERSISTS: LATENT SPATIAL FIELDS FOR THE RESIDUAL RISK OF ARMED CONFLICT AND SOCIAL UNREST

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[TODO: Abstract to be written after the application.]

1. Introduction. [TODO: Introduction (not drafted yet).]

2. Data.

2.1. *Conflict and unrest events.* Event data come from the Armed Conflict Location & Event Data Project (ACLED; Raleigh et al., 2010). ACLED records the date, location, actors and type of individual events of political violence and demonstrations, compiled from news media, local partners and other sources. We use events from 1 January 2018 to 31 December 2024 and group ACLED event types into two outcomes: *violent conflict* (battles, explosions/remote violence and violence against civilians) and *social unrest* (riots and protests). Events are aggregated to a regular $0.5^\circ \times 0.5^\circ$ longitude–latitude grid, the resolution of PRIO-GRID (Tollefsen, Strand and Buhaug, 2012), and to calendar weeks. For cell s and week t, let $N_{st}^{(j)}$ be the number of events of type j, and let

$$Y_{st} = \mathbf{1}\Big\{\textstyle\sum_{j\in\mathcal{J}} N_{st}^{(j)} > 0\Big\},$$

where $\mathcal{J}$ is the set of event types of the outcome being modelled. The two outcomes are analysed separately. Event weeks are rare. In 2023–2024, over the covered cell-weeks of Africa and the Middle East with complete history, [TODO: x]% of cell-weeks contain at least one conflict event and [TODO: x]% at least one unrest event. Over all covered cell-weeks worldwide the shares are [TODO: x]% and [TODO: x]%.

Figure 1 shows where and when events are recorded. Both outcomes are spatially concentrated. Conflict event weeks are clustered in a few regions, which are discussed in Section 5.3. Over the study period, [TODO: x]% of cells record at least one conflict event and [TODO: x]% at least one unrest event, and the 1% of cells with the most event weeks account for [TODO: x]% and [TODO: x]% of all event weeks. A small number of cells record events in more than half of all weeks, which is why recent history is such a strong predictor (Section 2.2). Unrest is more widespread, and its distribution follows population more closely. The weekly share of cells with an event varies between [TODO: x]% and [TODO: x]% for conflict and between [TODO: x]% and [TODO: x]% for unrest, without a strong trend over the period. The test period of 2023–2024 therefore does not differ markedly in overall frequency from the training period.

The grid consists of all 0.5° cells that intersect a country polygon of Natural Earth (1:10m), excluding cells without a terrestrial biome (rock and ice, and large lakes). Each cell is assigned to the country that contains its centroid. A coastal cell whose centroid falls in the sea is assigned to the nearest country. National indicators are attached through that country only,

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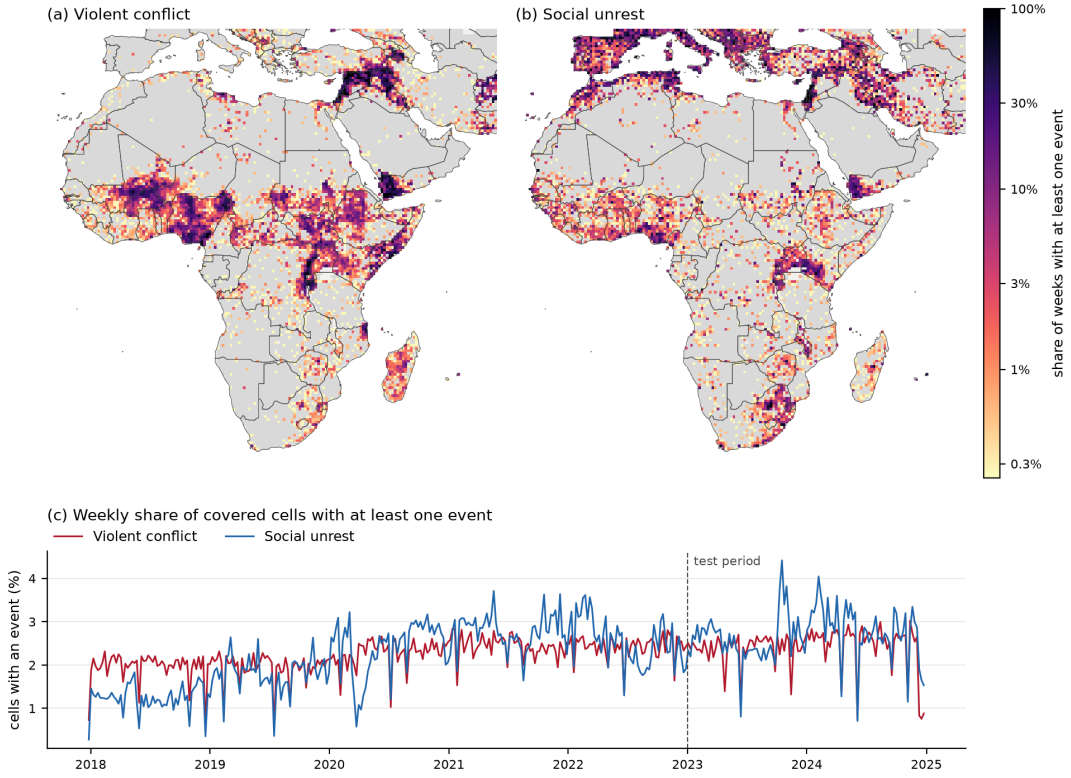


FIG 1. *ACLED events on the* 0.5° *grid, 2018–2024, over covered cell-weeks. (a, b) Share of weeks with at least one event in each cell, for violent conflict and social unrest (logarithmic colour scale); grey cells have no event week. (c) Weekly share of covered cells with at least one event. The dashed line marks the start of the test period.*

so every cell-week appears exactly once. The result is 59,344 cells observed for up to 366 weeks.

The table built by the original pipeline differs in one important respect. About 9% of its cell-weeks occur more than once, with multiplicities such as 2^k, 3^k and $2 \cdot 3^k$, the signature of many-to-many joins between the grid and polygons of countries, IPC regions and national indicators. These copies make up 50.5% of its 70.0 million rows, for 34.7 million distinct cell-weeks before the biome exclusion. The copies of a cell-week share its events but often disagree on the covariates. Among duplicated cell-weeks, the copies differ in the EPI and the liberal democracy index in 62% of cases, in GDP in 49%, in the IPC phases in about 25% and in the climate variables in 7–11%. We keep one copy of each cell-week. IPC phase populations are averaged over the copies. National indicators are re-attached for each country and year from the cells of that country that were not duplicated. GDP per capita is taken from the World Bank and is a national quantity. Section 5.1 describes the consequences of the duplication for the original analysis.

Two features of ACLED matter for modelling. First, geographic coverage expanded over time, and several countries entered the dataset after 2018. Cell-weeks in a country before its coverage start are *unobserved*, not event free, so they are removed rather than coded as zeros. Coverage start dates are taken from ACLED's country coverage table. Territories not listed there take the date of the country or region that contains them. Coverage removes [TODO: x] cell-weeks in [TODO: x] countries, and the event history is treated as incomplete in the first four covered weeks. Second, the probability that an event is recorded depends on media and partner density, so the outcome combines occurrence and reporting (Weidmann, 2016). The

two are not separately identified without auxiliary information. The latent fields of Section 3 absorb spatially smooth reporting differences together with unmeasured drivers of risk, and we interpret them accordingly.

2.2. *Recent event history.* For each cell and week we compute the number of events of each type in the four preceding weeks, $L_{st}^{(j)} = \sum_{l=1}^{4} N_{s,t-l}^{(j)}$, using only past weeks of the same cell. In the models of Section 3 the history enters through $h_{st} = \log\bigl(1 + \sum_{j \in \mathcal{J}} L_{st}^{(j)}\bigr)$ for the outcome being modelled, and through the corresponding quantity for the other outcome as an additional covariate. Cell-weeks whose four-week history is incomplete, at the start of the series and of each country's coverage, are dropped from the latent models.

2.3. *Environmental, socioeconomic and national covariates.* Covariates are listed in Table 1. They vary at three resolutions, which is central to the identification arguments of Section 3.6:

1. *cell–week*: daily minimum and maximum temperature, heat index and wet-bulb globe temperature (CHIRTS), precipitation (CHIRPS), $PM_{2.5}$ (CAMS) and a drought index (Global Drought Observatory);
2. *cell*: population (GHSL), river length and lake area (Natural Earth), malaria risk (Malaria Atlas Project), terrestrial biome (RESOLVE Ecoregions) and distance from the equator;
3. *country–year*: GDP per capita, access to electricity, the Environmental Performance Index (EPI), the V-Dem liberal democracy index, and per-capita greenhouse gas emissions, material footprint and ecological footprint.

Food insecurity, the population in each IPC phase, is available for subnational regions and periods and is apportioned to cells by population share.

The climate variables enter all models through their means over the four weeks before the current week, so that they precede the outcome. The boosting models use these means on their original scale, as in the original analysis. A raw temperature means different things in different climates. In the latent models, the five climate variables (maximum and minimum temperature, precipitation, drought and $PM_{2.5}$) therefore enter as standardised anomalies with respect to a cell- and calendar-month-specific climatology estimated from the training years 2018–2022,

$$z_{st} = \bigl(x_{st} - \bar{x}_{s,\mathrm{m}(t)}\bigr)/\mathrm{sd}_{s,\mathrm{m}(t)},$$

where $\mathrm{m}(t)$ is the calendar month of week t. Their coefficients are then identified from variation over time within cells and are not confounded by the persistent spatial pattern of the climatology. The cell's mean over the training years enters separately as a cell-level covariate. A five-year climatology is noisy, especially for precipitation, and a longer reference period would be preferable where available; the anomalies are therefore best read as departures from recent conditions.

All covariates of the latent models other than the event histories are standardised with training-period means and standard deviations. Missing values are set to zero, the training mean.

2.4. *Exploratory evidence of residual spatial dependence.* Before introducing the latent models, we check whether a flexible model of the covariates and recent history leaves spatial structure in its residuals. We take the XGBoost offset of Section 3.7, cross-fitted by country so that a cell's prediction never uses data from its own country, and compute the standardised cell residuals of Section 3.9 on the test years. In Africa and the Middle East, Moran's I of

TABLE 1
Covariates, resolution and source.

Covariate	Resolution	Source
Minimum / maximum temperature	cell–week	CHIRTS
Heat index, wet-bulb globe temperature	cell–week	CHIRTS
Precipitation	cell–week	CHIRPS
$PM_{2.5}$	cell–week	CAMS
Drought index	cell–week	Global Drought Observatory
Events of each type in previous 4 weeks	cell–week	ACLED
Population	cell	GHSL
River length, lake area	cell	Natural Earth
Malaria risk	cell	Malaria Atlas Project
Biome	cell	RESOLVE Ecoregions
Distance from the equator	cell	derived
Food insecurity (IPC phases 1–5)	subnational–period	IPC
GDP per capita	country–year	World Bank
Access to electricity	country–year	World Bank
Environmental Performance Index	country–year	Yale EPI
Liberal democracy index	country–year	V-Dem
GHG emissions, material and ecological footprint p.c.	country–year	OWID, GFN

these residuals is [TODO: x] for conflict and [TODO: x] for unrest. The residual correlation between neighbouring cells is [TODO: x] within countries and [TODO: x] across borders for conflict. The semivariogram is reported in Section 5.3. This residual structure motivates the latent fields of Section 3.

3. Methods.

3.1. *A latent Gaussian model for event occurrence.* Let $\mathcal{S}$ be the set of n grid cells, $c(s)$ the country and $b(s)$ the biome of cell s, and $t = 1, \dots, T$ the weeks. Conditionally on a set of latent effects, the outcomes are independent Bernoulli variables, $Y_{st} \mid \pi_{st} \sim \text{Bernoulli}(\pi_{st})$, with

$$\text{logit}\, \pi_{st} = \boldsymbol{x}_{st}^{\top}\boldsymbol{\beta} + \delta\, h_{st} + \sum_{k \in \mathcal{K}} z_{k,st}\, \beta_k(s) + \boldsymbol{z}_{c(s),t}^{\top}\boldsymbol{\alpha} + u_{c(s)} + w(s) + v(s,t). \tag{3.1}$$

Here h_{st} is the recent event history of Section 2.2, so δ measures state dependence (Section 3.4). The vector $\boldsymbol{x}_{st}$ holds the cell and cell-week covariates with fixed effects. In the application these are the other outcome's event history, population, malaria risk, the IPC phase 3+ population, river length and lake area, the climate anomalies and climatologies of Section 2.3, biome effects and year effects. The variables $z_{k,st}$, $k \in \mathcal{K}$, are climate anomalies whose effects $\beta_k(s)$ vary in space (Section 3.5); in the application $\mathcal{K}$ contains maximum temperature only. The vector $\boldsymbol{z}_{ct}$ holds the national indicators with effects $\boldsymbol{\alpha}$ (Section 3.6). The country effects $u_c \sim \mathcal{N}(0, \sigma_u^2)$ are independent, w is a Matérn field on the sphere (Section 3.2), and v is a space-time field (Section 3.3). The three latent terms capture persistent *recorded* risk at three scales. Each mixes occurrence with reporting (Section 2.1): national institutions and reporting regimes (u), regional hotspots that cross borders (w), and slowly evolving episodes (v). Missing covariate values are set to the training-period mean after standardisation.

Model (3.1) is a latent Gaussian model (Rue, Martino and Chopin, 2009). Stacking all latent quantities into $\boldsymbol{\xi} = (\boldsymbol{\beta}, \delta, \boldsymbol{\gamma}, \boldsymbol{\alpha}, \boldsymbol{u}, \boldsymbol{w}, \boldsymbol{\eta}, \boldsymbol{v})$, the linear predictor is $\boldsymbol{\eta} = \boldsymbol{o} + \boldsymbol{A}\boldsymbol{\xi}$ for a

TABLE 2
Latent blocks of model (3.1), their precision matrices and hyperparameters.

Block	Dimension	Precision	Hyperparameters
fixed effects $\boldsymbol{\beta}, \delta, \boldsymbol{\gamma}, \boldsymbol{\alpha}$	p	$10^{-4}\boldsymbol{I}$	–
country effects $\boldsymbol{u}$	n_c	$\sigma_u^{-2}\boldsymbol{I}$	σ_u
spatial field $\boldsymbol{w}$	n	$\boldsymbol{Q}_s(\rho_w, \sigma_w)$, eq. (3.3)	ρ_w, σ_w
SVC fields $\boldsymbol{\eta}_k$	n each	$\boldsymbol{Q}_s(\rho_k, \sigma_k)$	ρ_k, σ_k
space-time field $\boldsymbol{v}$	nT_v	$\boldsymbol{Q}_{\mathrm{AR}}(\phi) \otimes \boldsymbol{Q}_s(\rho_v, \sigma_v)$	ρ_v, σ_v, ϕ

sparse design matrix $\boldsymbol{A}$ and an optional offset $\boldsymbol{o}$, and $\boldsymbol{\xi} \mid \boldsymbol{\theta} \sim \mathcal{N}\{\boldsymbol{0}, \boldsymbol{Q}(\boldsymbol{\theta})^{-1}\}$ with a sparse, block-diagonal precision whose blocks are listed in Table 2 (Rue and Held, 2005). Fixed effects receive vague $\mathcal{N}(0, 10^4)$ priors and are integrated out together with the random effects.

3.2. *Matérn fields on the sphere via a finite-volume SPDE.* Following Lindgren, Rue and Lindström (2011), a Gaussian field w on $\mathbb{S}^2$ with Matérn-like covariance is defined as the stationary solution of

$$\tau\,(\kappa^2 - \Delta_{\mathbb{S}^2})\,w(s) = \mathcal{W}(s), \qquad s \in \mathbb{S}^2, \tag{3.2}$$

where $\Delta_{\mathbb{S}^2}$ is the Laplace–Beltrami operator and $\mathcal{W}$ is Gaussian white noise (see also Lindgren, Bolin and Rue, 2022). This is the $\alpha = 2$ case, which corresponds to Matérn smoothness $\nu = 1$ in the planar limit. The field then has practical range $\rho = \sqrt{8}/\kappa$ and marginal variance $\sigma^2 = 1/(4\pi\kappa^2\tau^2)$, provided ρ is small relative to the Earth's radius R. We parameterise the field by (ρ, σ), in kilometres and logit units.

The grid cells are natural control volumes, so instead of a triangulated mesh we discretise (3.2) by finite volumes on the longitude–latitude lattice itself (for finite-volume SPDE discretisations on regular grids, see Fuglstad et al., 2015). Integrating (3.2) over cell A_i and applying the divergence theorem with a two-point flux approximation gives

$$\tau\Big\{\kappa^2 a_i w_i + \sum_{j \sim i} \frac{\ell_{ij}}{d_{ij}}(w_i - w_j)\Big\} = \mathcal{W}(A_i) \sim \mathcal{N}(0, a_i),$$

where a_i is the area of cell i, $j \sim i$ ranges over the cells sharing an edge with i, ℓ_{ij} is the length of the shared edge and d_{ij} the great-circle distance between the centres. For a grid with spacings $\Delta\lambda, \Delta\varphi$ and centre latitude φ_i,

$$\begin{aligned} a_i &= R^2 \Delta\lambda\,\{\sin(\varphi_i + \Delta\varphi/2) - \sin(\varphi_i - \Delta\varphi/2)\}, \\ \ell_{ij}/d_{ij} &= \Delta\varphi/(\cos\varphi_i\,\Delta\lambda) \quad \text{(east–west neighbours)}, \\ \ell_{ij}/d_{ij} &= \cos\varphi_{ij}\,\Delta\lambda/\Delta\varphi \quad \text{(north–south neighbours)}, \end{aligned}$$

with φ_{ij} the latitude of the shared edge. Writing $\boldsymbol{C} = \mathrm{diag}(a_i)$ and $\boldsymbol{G}$ for the weighted graph Laplacian with edge weights ℓ_{ij}/d_{ij}, we get $\tau\boldsymbol{K}\boldsymbol{w} = \boldsymbol{C}^{1/2}\boldsymbol{\epsilon}$ with $\boldsymbol{K} = \kappa^2\boldsymbol{C} + \boldsymbol{G}$ and $\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$, so

$$\boldsymbol{w} \sim \mathcal{N}\{\boldsymbol{0}, \boldsymbol{Q}_s^{-1}\}, \qquad \boldsymbol{Q}_s(\rho, \sigma) = \tau^2 \boldsymbol{K}\boldsymbol{C}^{-1}\boldsymbol{K}, \qquad \log|\boldsymbol{Q}_s| = 2n\log\tau + 2\log|\boldsymbol{K}| - \textstyle\sum_i \log a_i. \tag{3.3}$$

$\boldsymbol{Q}_s$ has at most 13 non-zeros per row. The construction is the lumped-mass finite-volume analogue of the finite-element representation in Lindgren, Rue and Lindström (2011). Because it uses the physical cell geometry, it accounts for the shrinking east–west spacing towards the poles. It can also be restricted to land cells by deleting rows and columns, and it wraps around in longitude for global analyses. The continuum relations between (κ, τ) and (ρ, σ)

TABLE 3

Boundary treatments on a 60×30 lattice of 0.5° cells with a sea strait of 2, 4 or 6 cells (about 110, 220, 330 km): marginal variance of a coastal and of an inland land cell (nominal 1) and correlation between the two coastal cells facing each other across the strait, from 4,000 exact draws with $\rho = 400$ km. "Land only" removes the strait from the lattice; "sea = land" keeps it as ordinary cells without data.

Strait		Land only	Sea = land	$f = 0.1$	$f = 0.2$	$f = 0.5$
2 cells	var. coast	1.98	1.07	1.80	1.67	1.29
	var. inland	1.10	1.06	1.09	1.11	1.09
	corr. across	-0.01	0.49	-0.03	0.01	0.16
4 cells	var. coast	2.00	1.08	1.77	1.69	1.29
	var. inland	1.06	1.05	1.09	1.09	1.09
	corr. across	0.00	0.28	0.01	0.01	0.04
6 cells	var. coast	2.02	1.04	1.76	1.66	1.26
	var. inland	1.06	1.05	1.08	1.07	1.07
	corr. across	-0.02	0.14	0.01	0.01	0.01

hold on the lattice only when ρ spans several cells. A fitted range below two or three cells (100–150 km) indicates a field that the 0.5° lattice does not resolve; its $\hat{\sigma}$ is then not the marginal standard deviation, and we interpret such fields as cell-level effects.

REMARK 1. The discretisation imposes reflecting (Neumann) conditions on the boundary of the lattice, which roughly doubles the marginal variance of boundary cells (Section 4.2; see Khristenko et al., 2019, for the effect of boundary conditions on SPDE fields). For a land grid the boundaries are the coastlines. Extending the lattice into the sea removes the inflation but lets correlation leak across narrow seas, for example between Yemen and Somalia across the Gulf of Aden, which is substantively wrong for conflict risk. We therefore use the barrier model of Bakka et al. (2019): the SPDE is written as

$$u(s) - \nabla \cdot \left\{ \tfrac{r(s)^2}{8} \nabla u(s) \right\} = r(s) \sqrt{\pi/2}\, \sigma\, \mathcal{W}(s),$$

which reduces to (3.2) with $\kappa = \sqrt{8}/\rho$ when $r(s) \equiv \rho$. The lattice is extended by six rings of cells (3°) without data. Sea cells, identified from the Natural Earth country polygons, get range $r(s) = f\rho$. Land cells without data, such as those outside a regional analysis box, keep the range ρ, so that the field is not artificially decorrelated at the edge of the region. The finite-volume form is $\boldsymbol{K}\boldsymbol{u} = \boldsymbol{D}\boldsymbol{\epsilon}$ with $\boldsymbol{K} = \boldsymbol{C} + (\rho^2/8)\boldsymbol{G}_m$, where $\boldsymbol{G}_m$ has edge weights $2m_i m_j/(m_i + m_j) \cdot \ell_{ij}/d_{ij}$, $m_i = r_i^2/\rho^2$ (the harmonic mean, the usual two-point flux choice for a discontinuous coefficient), and $\boldsymbol{D} = \mathrm{diag}\{r_i \sigma (\pi a_i/2)^{1/2}\}$, so that $\boldsymbol{Q}_s = \boldsymbol{K}\boldsymbol{D}^{-2}\boldsymbol{K}$. Table 3 reports the marginal variance at the coast and the correlation across straits of two to six cells for several choices of f. With $f = 0.1$, the value of Bakka et al. (2019), the barrier range is about half a cell and the coastline behaves almost like a reflecting boundary. We use $f = 0.5$, which keeps the coastal variance within about 30% of nominal and the correlation across straits of four cells or more (about 220 km) below 0.04. Narrower straits, below about 110 km, are not resolved at this grid resolution.

3.2.0.1. Priors.. We use penalised-complexity priors (Simpson et al., 2017; Fuglstad et al., 2019): $P(\rho < \rho_0) = 0.05$ and $P(\sigma > \sigma_0) = 0.01$, with $\rho_0 = 100$ km (about two grid cells) and $\sigma_0 = 1$. The PC prior of Fuglstad et al. (2019) is derived for Matérn fields on $\mathbb{R}^2$; we use it as an approximation on the sphere, which is adequate for ranges far below R. The country effect has a PC prior with $P(\sigma_u > 1) = 0.01$.

3.3. *Space-time persistence.* The space-time term is a separable field that evolves as a first-order autoregression of Matérn innovations,

$$\boldsymbol{v}_\tau = \phi\, \boldsymbol{v}_{\tau-1} + \sqrt{1-\phi^2}\, \boldsymbol{\epsilon}_\tau, \qquad \boldsymbol{\epsilon}_\tau \sim \mathcal{N}\{\mathbf{0}, \boldsymbol{Q}_s(\rho_v, \sigma_v)^{-1}\},$$

(Cameletti et al., 2013), so that $\boldsymbol{v}$ has precision $\boldsymbol{Q}_{\mathrm{AR}}(\phi) \otimes \boldsymbol{Q}_s(\rho_v, \sigma_v)$, where $\boldsymbol{Q}_{\mathrm{AR}}$ is the tridiagonal precision of a unit-variance AR(1) process. To control cost, τ indexes blocks of Δ weeks, and $v(s,t) = v_{s,\lceil t/\Delta \rceil}$. We use $\Delta = 1$ and $\Delta = 4$ in the simulations and $\Delta = 8$ in the application. The field is defined on the observed land cells only, so it cannot link cells across the sea. The prior on $\operatorname{atanh}\phi$ is $\mathcal{N}(0, 1.5^2)$. Non-separable alternatives exist (Lindgren et al., 2024) but are not needed for the questions asked here.

3.4. *State dependence versus persistent heterogeneity.* Recent event counts dominated the importance rankings of the original boosting analysis (Menicali, Marcantonio and Castruccio, 2026). Part of that dominance was an artefact of duplicated rows (Section 5.2), but after correction recent history remains the strongest single predictor. This has several explanations, which model (3.1) separates only partially:

1. *state dependence*: recorded events raise the probability of further recorded events in the same place, through retaliation, mobilisation or the arrival of armed groups;
2. *persistent heterogeneity*: some places are simply riskier (w and v are large there), and past events are informative only as a proxy for that latent risk (Heckman, 1981);
3. *reporting dynamics*: media attention and reporting networks themselves persist, so a place that has been reported on is more likely to be reported on again, whatever happens there (Weidmann, 2016).

All three produce strong autocorrelation in Y_{st} and a large predictive contribution of h_{st}. In the conflict literature the distinction appears as the question of whether apparent contagion between neighbouring places reflects spillover or shared conditions (Buhaug and Gleditsch, 2008). Self-exciting point process models address it by decomposing intensity into a background and a triggering component (Mohler et al., 2011; Mohler, 2013; Porter and White, 2012; Clark and Dixon, 2018; Reinhart, 2018; Jun and Cook, 2024).

In model (3.1), δ is the effect of recent recorded history conditional on the latent persistence $w + v$, and it is identified only under the assumed latent dynamics. The space-time field v is resolved at steps of Δ weeks, and h_{st} covers the previous four weeks. When the two time scales are similar, separating δ from v relies on the AR(1) form, so we report δ for several Δ (Section 4.5). We therefore read $\hat{\delta}$ as conditional dependence of recorded events, not as a causal contagion effect. To limit the initial-conditions problem of dynamic random-effects models (Wooldridge, 2005), we drop weeks whose four-week history is incomplete (the first weeks of each cell and of each country's ACLED coverage). We also add each cell's history at its first usable week as a covariate, following Wooldridge (2005).

3.5. *Spatially varying climate effects.* To assess whether climate effects differ across biomes and how much variation remains within them, the effect of anomaly k is decomposed as

$$\beta_k(s) = \gamma_{k,b(s)} + \eta_k(s), \qquad \boldsymbol{\eta}_k \sim \mathcal{N}\{\mathbf{0}, \boldsymbol{Q}_s(\rho_k, \sigma_k)^{-1}\}, \tag{3.4}$$

a spatially varying coefficient model (Gelfand et al., 2003; Dambon, Sigrist and Furrer, 2021) whose mean is pooled by biome. The biome means $\gamma_{k,b}$ are fixed effects with the vague prior of Section 3.1. The field $\boldsymbol{\eta}_k$ has PC priors with $P(\rho_k < 100\,\mathrm{km}) = 0.05$ and $P(\sigma_k > 0.5) = 0.01$, and σ_k measures heterogeneity within biomes. The design of $\boldsymbol{\eta}_k$ is the sparse matrix with entry $z_{k,st}$ in row (s,t) and column s.

Three summaries of a biome-specific effect should be distinguished:

- *Fixed biome slopes.* A model with fixed biome-specific slopes and no $\boldsymbol{\eta}_k$ estimates an information-weighted average, $\sum_{s\in\mathcal{S}_b}\omega_s\beta_k(s)/\sum_{s\in\mathcal{S}_b}\omega_s$, with $\omega_s \approx \sum_t z_{k,st}^2\pi_{st}(1-\pi_{st})$. This average puts most weight on the high-risk cells of the biome.
- *Biome means.* Model (3.4) targets the biome mean $\gamma_{k,b}$, the mean of the process that generated the biome's coefficients. This is a superpopulation quantity. Because $\boldsymbol{\eta}_k$ is not constrained to average zero within biomes and biomes are large contiguous regions, $\gamma_{k,b}$ is separated from the biome average of $\boldsymbol{\eta}_k$ mainly when ρ_k is small relative to the size of the biome. When ρ_k is comparable to biome size, the separation rests partly on the prior. Its intervals account for the finite number of independent regions in each biome.
- *Median SHAP value.* The summary used in the SHAP-based analysis of Menicali, Marcantonio and Castruccio (2026), the median SHAP value within a biome, differs from both. For a feature with an additive effect, it equals $\mathrm{median}_{s\in\mathcal{S}_b}\{\beta_k(s)(z_{k,st}-\mathbb{E}z_k)\}$, which reflects where the biome's covariate values lie relative to the global mean as much as the strength of the effect. When the effect is a biome-by-feature interaction, SHAP also splits the interaction between the feature and biome (Kumar et al., 2020; Aas, Jullum and Løland, 2021), so that, for example, a pure interaction $g(b)z$ contributes $g(b)(z-\mathbb{E}z)/2$ to the feature's SHAP value.

3.6. *National indicators and spatial confounding.* National indicators $\boldsymbol{z}_{ct}$, such as GDP per capita, the EPI or the ecological footprint, take one value per country and year. We decompose each into a between-country component, its country mean over the training years $\bar{z}_c$, and a within-country component $z_{ct}-\bar{z}_c$, with separate coefficients (a Mundlak decomposition). Year effects absorb trends common to all countries. The between coefficients are identified from contrasts between the n_c countries only, so their effective sample size is of order n_c, not the number of cell-weeks. Modelling cell-weeks as conditionally independent, or bootstrapping spatial blocks that cut across countries, therefore produces intervals that are far too narrow. The within coefficients rest on the few years of change within each country.

If the indicators are associated with unobserved country-level risk, so that $\mathbb{E}\{u_c+\bar{w}_c \mid \bar{z}_c\}\neq 0$ with $\bar{w}_c$ the country average of w, then the between coefficients are confounded. An indicator of environmental performance that tracks state capacity, or reporting capacity, is the leading example. This is the spatial confounding problem (Reich, Hodges and Zadnik, 2006; Hodges and Reich, 2010; Paciorek, 2010). Adding a spatial random effect changes the estimate of a spatially smooth covariate's coefficient, and in general neither the estimate with the random effect nor the one without it is unbiased for a causal effect (Khan and Calder, 2022; Zimmerman and Ver Hoef, 2022). Recent work has made this more precise. Generalised least squares with a spatial random effect can be consistent when the covariate has non-spatial variation (Gilbert, Ogburn and Datta, 2025). Confounding in the data-generating process should be distinguished from the behaviour of the analysis model (Khan and Berrett, 2026). The bias depends on how the covariate and the confounder are spread across spatial scales (Guan et al., 2023; Dupont, Marques and Kneib, 2026). A country-constant indicator has no within-country spatial variation, so these results give little reassurance here.

We report the following estimators and treat their disagreement as a measure of how fragile the national-level conclusions are:

1. model (3.1) without $\boldsymbol{u}$ and $\boldsymbol{w}$;
2. model (3.1) with $\boldsymbol{u}$;
3. model (3.1) with $\boldsymbol{u}$ and $\boldsymbol{w}$;
4. a *spatial+* estimator (Dupont, Wood and Augustin, 2022), which is (iii) with each between component replaced by its residual after an area-weighted projection, over the observed land cells, on the m smoothest eigenvectors of $\boldsymbol{G}\boldsymbol{e}=\lambda\boldsymbol{C}\boldsymbol{e}$.

These eigenvectors are the lattice analogue of the thin-plate eigenvectors of Dupont, Wood and Augustin (2022). We take m as the number of eigenvalues below $\kappa_w^2 = 8/\hat{\rho}_w^2$, the eigenvectors that vary on scales longer than the fitted range of w, and report $m = 10$ and $m = 20$ as sensitivity checks. A country-constant indicator varies by steps at borders, so the residual keeps the between-country contrasts that are not smooth at the scale of $\hat{\rho}_w$. The restricted spatial regression of Reich, Hodges and Zadnik (2006) is not used, because its intervals are known to be anticonservative (Hanks et al., 2015; Khan and Calder, 2022).

3.7. *Hybrid models: boosting offsets with latent effects.* To retain the flexibility of gradient boosting for the many weakly informative covariates, we also fit a semiparametric version of (3.1). Its fixed part is replaced by the logit $\hat{f}(\boldsymbol{x}_{st})$ of an XGBoost classifier (Chen and Guestrin, 2016), which enters as an offset:

$$\operatorname{logit} \pi_{st} = \hat{f}^{(-k(s))}(\boldsymbol{x}_{st}) + \beta_0 + u_{c(s)} + \omega(s), \tag{3.5}$$

where ω is either the Matérn field w or independent cell effects. The offset is *cross-fitted by country*. Countries are assigned at random to $K = 5$ folds. For each fold k, $\hat{f}^{(-k)}$ is trained on the 2018–2022 rows of the other folds, with its number of trees chosen by early stopping on 2022 after training on 2018–2021. It then provides the offset for every row of fold k, in both the training and test periods. Trees that receive national indicators can identify countries, and through population, biome and distance from the equator, approximate locations. Cross-fitting by country ensures that the offset of a cell was learned without any data from its country, so that u and ω can be estimated on top of it without double counting. The price is that the offset is weaker than a boosting model trained on all locations. Comparisons with the latter are therefore made on identical test rows and include location-aware boosting baselines (Section 3.9). The offset uses the same rows, features and hyperparameters as the corrected boosting model. It is trained without class weights, so it is on the log-odds scale. Inference for u and ω is conditional on the offset, whose estimation uncertainty is not propagated. Cross-fitting is used here in the loose sense of sample splitting, not with an orthogonal score as in Chernozhukov et al. (2018).

Model (3.5) differs from GPBoost (Sigrist, 2022, 2023), which alternates boosting steps with latent Gaussian model updates: here the boosting fit is fixed before the latent model is fitted. This makes the model cheap enough for the global data and allows the same Laplace machinery to be used. Random forests with spatially correlated errors (Saha, Basu and Datta, 2023) are a further alternative.

The boosting analysis of Menicali, Marcantonio and Castruccio (2026) trained XGBoost with positives weighted by $\omega_+ = (n_0/n_1)^{1/2}$. Under the logistic loss this shifts the fitted log-odds by $\log \omega_+$ (King and Zeng, 2001), so its predictions are prior-corrected, $\hat{f} \leftarrow \hat{f} - \log \omega_+$, before probabilistic scoring. Class reweighting does not improve discrimination and miscalibrates probabilities (van den Goorbergh et al., 2022).

3.8. *Inference.* For fixed hyperparameters $\boldsymbol{\theta}$, the posterior mode $\boldsymbol{\xi}^* = \arg\max_{\boldsymbol{\xi}} \{\log p(\boldsymbol{y} \mid \boldsymbol{\xi}) - \frac{1}{2}\boldsymbol{\xi}^\top \boldsymbol{Q}\boldsymbol{\xi}\}$ is found by damped Newton iterations,

$$\boldsymbol{\xi} \leftarrow \boldsymbol{\xi} + \lambda\,(\boldsymbol{Q} + \boldsymbol{A}^\top \boldsymbol{W} \boldsymbol{A})^{-1}\{\boldsymbol{A}^\top(\boldsymbol{y} - \boldsymbol{\pi}) - \boldsymbol{Q}\boldsymbol{\xi}\}, \qquad \boldsymbol{W} = \operatorname{diag}\{\pi_{st}(1 - \pi_{st})\},$$

with step halving. The Laplace approximation to the marginal likelihood is

$$\log \tilde{p}(\boldsymbol{y} \mid \boldsymbol{\theta}) = \log p(\boldsymbol{y} \mid \boldsymbol{\xi}^*) - \tfrac{1}{2}\boldsymbol{\xi}^{*\top}\boldsymbol{Q}\boldsymbol{\xi}^* + \tfrac{1}{2}\log|\boldsymbol{Q}| - \tfrac{1}{2}\log|\boldsymbol{Q} + \boldsymbol{H}|, \qquad \boldsymbol{H} = \boldsymbol{A}^\top \boldsymbol{W}^* \boldsymbol{A}. \tag{3.6}$$

The hyperparameters are set to $\hat{\boldsymbol{\theta}} = \arg\max\{\log \tilde{p}(\boldsymbol{y} \mid \boldsymbol{\theta}) + \log p(\boldsymbol{\theta})\}$ by L-BFGS-B with finite-difference gradients on the internal scale (logarithms of ranges and standard deviations,

$\operatorname{atanh}\phi$). Each evaluation warm-starts Newton at the previous mode. Given $\hat{\boldsymbol{\theta}}$, the latent posterior is approximated by $\mathcal{N}\{\boldsymbol{\xi}^*, (\boldsymbol{Q}+\boldsymbol{H})^{-1}\}$. Samples reuse the factorisation of $\boldsymbol{Q}+\boldsymbol{H}$ already computed at the mode, through the perturbation identity

$$\boldsymbol{\xi} = \boldsymbol{\xi}^* + (\boldsymbol{Q}+\boldsymbol{H})^{-1}\big(\boldsymbol{R}^\top \boldsymbol{\epsilon}_1 + \boldsymbol{A}^\top \boldsymbol{W}^{*1/2}\boldsymbol{\epsilon}_2\big),$$

where $\boldsymbol{R}^\top\boldsymbol{R} = \boldsymbol{Q}$ is block-wise explicit, for instance $\boldsymbol{R} = \boldsymbol{D}^{-1}\boldsymbol{K}$ for a Matérn block and a Kronecker product of a bidiagonal AR(1) factor with that matrix for $\boldsymbol{v}$. This is an empirical Bayes version of INLA (Rue, Martino and Chopin, 2009) and is similar in spirit to TMB (Kristensen et al., 2016). Laplace approximations are generally accurate for binary GLMMs but can understate uncertainty when there are few observations per latent variable and events are rare (Fong, Rue and Wakefield, 2010; Ferkingstad and Rue, 2015). The intervals also ignore the uncertainty in $\hat{\boldsymbol{\theta}}$. Section 4.2 therefore checks coverage at the event prevalence of the data, and Section 4.5 checks it for the space-time model, where each latent value is informed by only Δ Bernoulli observations.

3.8.0.1. Computation.. All matrices are sparse. Every Newton step and every log-determinant requires a sparse factorisation of $\boldsymbol{Q}+\boldsymbol{H}$, whose sparsity pattern does not depend on $\boldsymbol{\theta}$. We use a sparse LU factorisation in symmetric mode (SuperLU) with a nested-dissection ordering (Karypis and Kumar, 1998), computed once and reused. For the space-time model this ordering reduced factorisation time by a factor of 7 to 14 compared with multiple minimum degree on $\boldsymbol{A}^\top + \boldsymbol{A}$. Since $\boldsymbol{H} = \boldsymbol{A}^\top\boldsymbol{W}\boldsymbol{A}$ aggregates over weeks, the cost of the spatial and SVC blocks is driven by the number of cells, not by the number of cell-weeks. A global 0.5° land grid gives about 6×10^4 latent values per spatial field. The space-time block has nT_v latent values, which is why it is fitted on a regional subset. The fixed-effect and country columns add dense rows to $\boldsymbol{Q}+\boldsymbol{H}$; with tens of fixed effects their cost is small.

3.8.0.2. Prediction.. For a new cell-week with design row $\boldsymbol{a}$ and offset o, the posterior predictive probability is $\mathbb{E}\{\operatorname{expit}(o+\boldsymbol{a}^\top\boldsymbol{\xi})\} \approx \operatorname{expit}\{m/\sqrt{1+\pi s^2/8}\}$, where m and s^2 are the posterior mean and variance of $o+\boldsymbol{a}^\top\boldsymbol{\xi}$; s^2 is estimated from posterior samples. Test years beyond the training period take the year effect of the last training year.

3.9. *Diagnostics and evaluation.*

3.9.0.1. Residual diagnostics.. Binary residuals are uninformative one at a time, so we aggregate over weeks within cells. The standardised cell residual

$$r_s = \sum_t (y_{st} - \hat{\pi}_{st}) \Big/ \Big\{\sum_t \hat{\pi}_{st}(1-\hat{\pi}_{st})\Big\}^{1/2}$$

is approximately standard normal, and independent across cells, when the model is correct, the outcomes are conditionally independent and $\hat{\pi}_{st}$ does not use the cell's own outcomes. We therefore apply these diagnostics to test-period residuals, and to training-period residuals of the cross-fitted offset only. Residuals of in-sample latent-model fits are shrunk and spatially dependent through the smoother. We report:

- Moran's I (Moran, 1950) of $\{r_s\}$ for rook adjacency, with a permutation p-value from $B = 999$ permutations, which is bounded below by $1/(B+1)$;
- the empirical semivariogram of $\{r_s\}$ against great-circle distance;
- the correlation of r_s between neighbouring cells in the same country and across a border.

A larger within-country correlation points to effects at the country level, of occurrence or of reporting.

3.9.0.2. Out-of-sample evaluation.. Models are trained on 2018–2022 and evaluated on 2023–2024, matching the original analysis. All models are compared on identical test rows. We report the logarithmic score and the Brier score, both strictly proper (Gneiting and Raftery, 2007), and the area under the precision–recall curve, computed as average precision (Boyd, Eng and Page, 2013). The latter is more informative than the ROC curve at the low prevalence of the data (Saito and Rehmsmeier, 2015). Paired differences in log score are reported with standard errors clustered by cell. Since prediction in the same places is dominated by persistence (Mueller and Rauh, 2022), every model is compared with a persistence baseline that ranks cell-weeks by the number of events in the previous four weeks. We also compare it with a *location-aware* boosting baseline that adds each cell's training-period event rate as a feature, computed leaving out the row's own year. This baseline is the boosting counterpart of a cell intercept, and the latent fields have to improve on it to be worth their cost.

4. Simulation studies. The simulations address five questions that the application raises but cannot settle on its own:

(S0) whether the finite-volume SPDE and the empirical Bayes Laplace approximation recover the field and give calibrated intervals, including at the event prevalence of the data;
(S1) whether boosting leaves residual spatial structure, and whether modelling it improves forecasts for the same places beyond a location-aware boosting baseline;
(S2) how national indicators behave under spatial confounding;
(S3) whether state dependence can be separated from persistent heterogeneity;
(S4) whether biome-specific climate effects are recovered better by spatially varying coefficients than by SHAP summaries.

All code is in the supplementary material.

4.1. *Data-generating process.* The simulator mimics the structure of the ACLED grid. The domain is a 40×30 block of $0.5°$ cells between $10°$E and $30°$E and between $5°$S and $10°$N (1,200 cells of about 55 km), with 104 training weeks, 52 test weeks and 12 burn-in weeks. Countries are the Voronoi cells of randomly placed seeds (14 countries; 25 in S2). Four biomes are latitude bands with boundaries perturbed by a smooth random field, ordered from "cold" (biome 1) to "tropical" (biome 4). Population is a standardised Matérn field (range 250 km) plus white noise. Temperature and precipitation anomalies are standardised AR(1) processes in time (coefficients 0.6 and 0.3) of Matérn fields with ranges of 800 and 500 km. A raw temperature is also generated, as a smooth climatology that rises from biome 1 to biome 4, plus a seasonal cycle and 2.5 times the anomaly. Three national covariates Z_0, Z_1, Z_2 are standard normal across countries. Z_1 and Z_2 have no effect but, like the national indicators in the data, allow trees to identify countries. Outcomes are generated sequentially from

(4.1)
$$\operatorname{logit} \pi_{st} = \beta_0 + \beta_{\text{pop}} \text{pop}_s + \delta \log\Big(1 + \textstyle\sum_{l=1}^{4} y_{s,t-l}\Big) + \beta(s)\, \text{temp}_{st} + \beta_{\text{prec}} \text{prec}_{st} + \alpha Z_{0,c(s)} + u_{c(s)} + w(s) + v(s,t),$$

with $\beta(s) = \gamma_{b(s)} + \eta(s)$ and $u_c \sim \mathcal{N}(0, 0.3^2)$. To avoid generating data from the model used for fitting, w, η and the innovations of v are drawn from Gaussian processes with the continuous Matérn covariance ($\nu = 1$) in great-circle distance, evaluated at the cell centres. They are not drawn from the lattice GMRF of Section 3.2. Unless stated otherwise, $\beta_0 = -4$, $\beta_{\text{pop}} = 0.5$, $\delta = 0.6$, $\boldsymbol{\gamma} = (-0.3, -0.1, 0.1, 0.3)$, $\eta \equiv 0$, $\beta_{\text{prec}} = -0.15$, $\alpha = 0$, $\sigma_w = 1$, $\rho_w =$ 400 km and $v \equiv 0$. The resulting prevalence of event weeks is about 4%. Spatial confounding (S2) is induced by setting $Z_{0,c} = \varrho\, \tilde{\ell}_c + (1 - \varrho^2)^{1/2} \epsilon_c$, where $\tilde{\ell}_c$ is the standardised country average of $u_c + w(s)$ and $\epsilon_c \sim \mathcal{N}(0, 1)$.

TABLE 4

S0: accuracy of the finite-volume SPDE field and of the empirical Bayes Laplace approximation, with fields drawn from the continuous Matérn process (50 replicates per setting). Mean over replicates, with the SD across replicates in parentheses, except for the coverage of b_1, where it is the Monte Carlo standard error.

Quantity	common ($b_0 = -2$, 50 reps)	rare ($b_0 = -4.5$, 50 reps)
Prevalence of event weeks	0.161 (0.030)	0.019 (0.005)
$\hat{\rho}_w$ (km) [truth 400]	420 (40)	409 (51)
$\hat{\sigma}_w$ [truth 1]	0.884 (0.064)	0.875 (0.075)
$\hat{b}_1$ [truth 0.5]	0.494 (0.016)	0.502 (0.021)
Coverage of 95% interval for b_1	0.88 (0.05)	0.96 (0.03)
Coverage of 95% intervals for $b_0 + w(s)$ (share of cells)	0.948 (0.009)	0.943 (0.017)
$\mathrm{cor}(\hat{w}, w)$	0.954 (0.008)	0.914 (0.018)
Fit time (s)	2.4 (0.7)	3.4 (0.8)
Marginal variance of w, interior cells [nominal 1]	1.07	–
Marginal variance of w, edge cells	2.02	–

4.1.0.1. Boosting.. Two versions of XGBoost are used, both with the hyperparameters of the original analysis (Menicali, Marcantonio and Castruccio, 2026): depth 5, learning rate 0.05, row and column subsampling 0.8, minimum child weight 30 and L_2 penalty 5. The *original-like* version follows the original procedure. Positives are weighted by $(n_0/n_1)^{1/2}$. The number of trees is chosen by early stopping on the last 26 training weeks, after which the model is refitted on all training weeks. Its log-odds are prior-corrected before scoring (Section 3.7). The *unweighted* version uses 300 trees without class weights. The features are population, distance from the equator, biome, the two anomalies, Z_0–Z_2 and the four-week event count. As in the original analysis, they include no coordinates.

4.2. *S0: accuracy of the field representation and of the Laplace engine.* We draw w on the 40×30 domain from the continuous Matérn process with $\rho_w = 400$ km and $\sigma_w = 1$, and Bernoulli outcomes from $\mathrm{logit}\,\pi_{st} = \beta_0 + 0.5\,x_{st} + w(s)$ with $x_{st} \sim \mathcal{N}(0,1)$, in two settings with 50 replicates each. In the *common* setting, $\beta_0 = -2$ with 40 weeks (prevalence about 15%). In the *rare* setting, $\beta_0 = -4.5$ with 104 weeks (prevalence about 2%, as in the data). We also report the marginal variance of the lattice field itself, from 2,000 exact draws, in interior cells and at the edge of the domain. Table 4 summarises the results.

The lattice field has marginal variance 1.07 in cells at least eight cells (about 440 km) from the edge of the domain, against a nominal 1, and 2.02 at the edge. This is the doubling induced by the reflecting boundary that motivates the barrier model of Remark 1. Although the data are not generated from the lattice model, the range is recovered with little bias in both settings ($\hat{\rho}_w = 420$ and 409 km against 400). The standard deviation is underestimated by about 12% ($\hat{\sigma}_w = 0.88$ against 1). This reflects both the discretisation and the known downward bias of empirical Bayes variance estimates in binary models. The posterior mean of w has correlation 0.95 with the truth in the common setting and 0.91 in the rare one. Plug-in 95% intervals cover b_1 in 88% of replicates in the common setting and 96% in the rare one (Monte Carlo SE about 4.6% and 2.8%). They cover $\beta_0 + w(s)$ for 94.8% and 94.3% of cells. Ignoring the uncertainty in $\boldsymbol{\theta}$ therefore costs little here, including at the prevalence of the data. A fit takes 2–4 seconds on one core.

4.3. *S1: residual structure after boosting.* Data are generated from (4.1) with the default parameters. All models are trained on the first 104 weeks and evaluated on the following 52 weeks at the same cells, as in the original analysis. We compare the true probabilities (oracle); the original-like XGBoost; the same model with each cell's training event rate as an additional

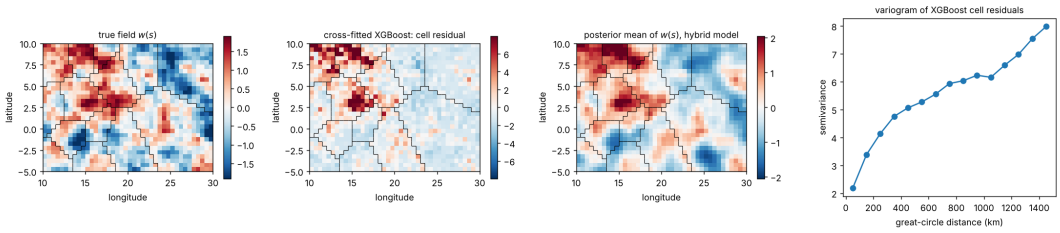


FIG 2. *S1, one replicate. From left: the true field w; standardised cell residuals of the XGBoost model cross-fitted by country (training period); posterior mean of w in the hybrid model (3.5); semivariogram of the XGBoost cell residuals. Colour scales are symmetric and capped at the 98th percentile of the absolute values. Lines are country borders.*

feature, computed leaving out the row's own 52-week period (the location-aware baseline of Section 3.9); the unweighted XGBoost cross-fitted by country; the hybrid model (3.5) with independent cell effects and with the Matérn field; the parametric model (3.1) with country effects and field; and the persistence baseline. Table 5 reports out-of-sample scores over 20 replicates. Table 6 reports Moran's I of the training-period cell residuals of the two XGBoost models and the recovery of the field. Figure 2 shows one replicate.

Three findings stand out. First, boosting leaves strong spatial structure in its residuals. Moran's I of the training-period cell residuals is 0.36 (SD 0.05 across replicates) for the original-like XGBoost, which is trained on the same locations, and 0.64 for the version cross-fitted by country. Under independence it is about -0.001. The trees recover part of the field through country-identifying national covariates, so in-sample residuals understate the spatial structure. The cross-fitted offset alone predicts poorly (AUC-PR 0.10 against 0.17), as it must, since it ignores everything that is specific to a country.

Second, the field estimated on top of the cross-fitted offset recovers the truth (correlation between $\hat{w}$ and w 0.85), with $\hat{\rho}_w = 409$ km against 400 and $\hat{\sigma}_w = 0.83$ against 1.

Third, modelling the latent structure improves forecasts for the same places. Differences below are paired by replicate, with standard errors over the 20 replicates.

- *Against the original-like XGBoost.* The hybrid model with the field improves the log score in all 20 replicates (mean difference -0.0027, SE 0.0002) and AUC-PR in all 20 ($+0.011$, SE 0.001).
- *Field against independent cell effects.* The field beats independent cell effects in every replicate (log score -0.0013, SE 0.0001; AUC-PR $+0.003$, SE 0.0003): smoothing across neighbouring cells helps.
- *Against the location-aware baseline.* The location-aware boosting baseline is a strong competitor. It improves on the original-like XGBoost about as much as the hybrid model does. Against it, the hybrid model with the field has a better log score in 16 of 20 replicates (mean difference -0.0011, SE 0.0004) but a similar AUC-PR (-0.001, SE 0.001). The hybrid model with independent cell effects is no better than the baseline.
- *Parametric model.* The correctly specified parametric model beats the baseline in every replicate in both log score and AUC-PR.

The gain from the latent field is therefore mostly in calibration rather than in ranking, once boosting is given each cell's own history. The persistence baseline reaches an AUC-PR of 0.095, against 0.20 for the true probabilities. In these simulations persistence is a weaker benchmark than in the data, where some cells record events almost every week.

4.4. *S2: national indicators under spatial confounding.* Three scenarios with 25 countries and 30 replicates each are considered: no confounding with $\alpha = 0$ ($\varrho = 0$); confounding

TABLE 5

S1: out-of-sample scores on the 52 test weeks (mean over 20 replicates, Monte Carlo standard error in parentheses). Lower is better for the log and Brier scores. The last two columns give the number of replicates in which the model has a better log score than (a) the original-like XGBoost and (b) the location-aware XGBoost with the cell training rate.

Model	Log score	Brier	AUC-PR	AUC-ROC	< (a)	<
True probabilities	0.1343 (0.0070)	0.0334 (0.0022)	0.200 (0.015)	0.805 (0.004)	20/20	20
XGBoost as in the original analysis	0.1401 (0.0072)	0.0341 (0.0023)	0.172 (0.015)	0.767 (0.006)	–	2/
XGBoost + cell training rate	0.1385 (0.0070)	0.0338 (0.0022)	0.185 (0.015)	0.781 (0.005)	18/20	
XGBoost, cross-fitted by country	0.1537 (0.0085)	0.0360 (0.0026)	0.101 (0.011)	0.670 (0.008)	0/20	0/
Cross-fitted XGBoost + u + iid cell effects	0.1387 (0.0071)	0.0339 (0.0023)	0.181 (0.015)	0.778 (0.005)	19/20	6/
Cross-fitted XGBoost + $u + w$	0.1374 (0.0071)	0.0339 (0.0023)	0.184 (0.015)	0.789 (0.005)	20/20	16
Parametric model with $u + w$	0.1365 (0.0071)	0.0336 (0.0022)	0.189 (0.015)	0.792 (0.005)	20/20	20
Persistence baseline	–	–	0.095 (0.010)	–	–	

TABLE 6

S1: residual diagnostics and recovery of the field (mean and SD over 20 replicates).

Quantity	Mean	SD
Moran's I of cell residuals, XGBoost as in the original	0.360	0.050
Moran's I of cell residuals, cross-fitted XGBoost	0.637	0.067
$\mathrm{cor}(\hat{w}, w)$, hybrid with $u + w$	0.849	0.044
$\hat{\rho}_w$ (km) [truth 400]	408.726	49.880
$\hat{\sigma}_w$ [truth 1]	0.829	0.069

with $\alpha = 0$ ($\varrho = 0.8$); and confounding with a true effect $\alpha = -0.3$. We compare estimators (i)–(iv) of Section 3.6, with spatial+ using $m = 10$ eigenvectors, and the XGBoost summary of the original analysis. For the latter, the original-like XGBoost is fitted to a fixed 30% subsample of the training data. The effect of Z_0 is summarised by the least-squares slope of its SHAP values on Z_0. Its intervals come from 40 replicates of a spatial block bootstrap with 2.5° blocks (48 blocks on the domain), each refitted with the same number of trees and class weight. The reported estimate is the median over bootstrap replicates, as in the original analysis.

Table 7 reports the results. Without confounding, all estimators are centred on zero, but intervals that ignore the country-level structure of the covariate are badly anticonservative. Logistic regression reports a standard error of 0.016 against an empirical SD of 0.107. It covers in 23% of replicates and declares a non-existent effect significant in 77%. Country effects alone restore nominal coverage. The block-bootstrap summary of the original analysis is closer to nominal than logistic regression, but its 95% interval still excludes zero in 20% of replicates, and its interquartile range, the criterion used in the original analysis, in 50%, although Z_0 has no effect.

Under confounding with $\alpha = 0$, no estimator is unbiased:

- logistic regression and the country-effects model report $\hat{\alpha}$ of 0.47 and 0.49 and always reject;
- the spatial field removes about 60% of the bias ($\hat{\alpha} = 0.20$), and spatial+ removes about 70% (0.15, coverage 0.60);
- XGBoost reports a SHAP slope of 0.33, with an interquartile range that excludes zero in every replicate.

With a true protective effect ($\alpha = -0.3$), logistic regression, the country-effects model and XGBoost report effects of the *wrong sign* ($+0.22$, $+0.24$ and $+0.14$). The XGBoost in-

TABLE 7

S2: estimation of the effect α of a national indicator (30 replicates per scenario). Coverage of nominal 95% intervals, and the proportion of replicates in which the interval excludes zero. †Least-squares slope of SHAP values on Z_0, with the 95% block-bootstrap percentile interval and interquartile range.

Estimator	Mean	Emp. SD	Mean SE	Coverage	Reject $\alpha = 0$
No confounding, $\alpha = 0$					
Logistic regression	-0.007	0.107	0.016	0.23	0.77
+ country effects u	0.016	0.109	0.123	0.97	0.03
+ spatial field w	0.002	0.065	0.052	0.93	0.07
+ u + w	0.000	0.063	0.081	1.00	0.00
spatial+ with u + w	0.002	0.072	0.082	0.97	0.03
XGBoost SHAP slope†	-0.006	0.063	0.044	0.80	0.20
Confounding, $\alpha = 0$					
Logistic regression	0.465	0.115	0.018	0.00	1.00
+ country effects u	0.493	0.098	0.078	0.00	1.00
+ spatial field w	0.200	0.098	0.056	0.30	0.70
+ u + w	0.200	0.095	0.077	0.33	0.67
spatial+ with u + w	0.153	0.092	0.079	0.60	0.40
XGBoost SHAP slope†	0.332	0.132	0.056	0.00	1.00
Confounding, $\alpha = -0.3$					
Logistic regression	0.219	0.114	0.017	0.00	1.00
+ country effects u	0.235	0.104	0.079	0.00	0.80
+ spatial field w	-0.090	0.100	0.056	0.20	0.40
+ u + w	-0.091	0.098	0.077	0.37	0.33
spatial+ with u + w	-0.130	0.095	0.078	0.50	0.40
XGBoost SHAP slope†	0.138	0.091	0.049	0.00	0.87

terquartile range excludes zero in 97% of replicates. The spatial estimators recover the sign but only a third to a half of the magnitude (-0.09 to -0.13).

The lesson for the national sustainability indicators is that the data can show whether an association survives spatial adjustment, but not its causal size. A large SHAP contribution of a national indicator, stable across block-bootstrap replicates, is fully compatible with no effect at all.

The block bootstrap resamples spatial blocks, but Z_0 varies only between countries, and a country spans several blocks. The bootstrap therefore does not reproduce the sampling variability of a country-level effect, whose effective sample size is the number of countries (Section 3.6).

4.5. *S3: state dependence versus persistent heterogeneity.* Three data-generating processes on a smaller 16×14 domain (224 cells, 6 countries, 104 training weeks, 20 replicates each) produce autocorrelation in Y_{st} for different reasons:

- *state dependence*: $\delta = 1$ and a weak field ($\sigma_w = 0.3$);
- *heterogeneity*: no state dependence ($\delta = 0$), a strong static field ($\sigma_w = 1.3$) and a space-time AR(1) field ($\sigma_v = 0.7$, $\phi = 0.9$ per week, $\rho_v = 300$ km);
- *mixed*: $\delta = 0.5$, $\sigma_w = 1$ and a space-time field with $\sigma_v = 0.5$ and $\phi = 0.9$.

We fit model (3.1) with lags only, with lags and w, and with lags, w and v. The space-time field v is resolved at four-week steps ($\Delta = 4$) and at weekly steps ($\Delta = 1$, the resolution of the data-generating process). We also report the share of total mean absolute SHAP value that the unweighted XGBoost assigns to the lag feature. Results are in Table 8.

TABLE 8

S3: estimation of the state-dependence parameter δ (20 replicates per scenario), and the share of the XGBoost mean absolute SHAP value assigned to the lag feature.

Estimator	Mean $\hat{\delta}$	Emp. SD	Mean SE	Coverage
State dependence ($\delta = 1$)				
Lags only	1.263	0.107	0.136	0.60
Lags + w	1.032	0.131	0.139	0.90
Lags + w + v (4-week steps)	0.990	0.135	0.140	0.95
Lags + w + v (weekly steps)	0.997	0.143	0.140	0.95
XGBoost: share of mean |SHAP| for lags	0.108	0.034	–	–
Heterogeneity ($\delta = 0$)				
Lags only	1.614	0.386	0.099	0.00
Lags + w	0.464	0.093	0.111	0.00
Lags + w + v (4-week steps)	0.045	0.152	0.119	0.90
Lags + w + v (weekly steps)	0.011	0.162	0.120	0.90
XGBoost: share of mean |SHAP| for lags	0.094	0.038	–	–
Mixed ($\delta = 0.5$)				
Lags only	1.659	0.351	0.106	0.00
Lags + w	0.782	0.111	0.115	0.25
Lags + w + v (4-week steps)	0.545	0.152	0.120	0.85
Lags + w + v (weekly steps)	0.535	0.172	0.120	0.80
XGBoost: share of mean |SHAP| for lags	0.116	0.044	–	–

Under state dependence, ignoring the latent field inflates $\hat{\delta}$ by about a quarter (1.26; coverage 0.60), and all latent specifications are approximately unbiased (0.99–1.03) with coverage of 0.90–0.95. Under heterogeneity the contrast is much sharper. With lags only, $\hat{\delta} = 1.61$ although the true value is zero. Adding the static field reduces the estimate to 0.46, but no interval covers zero: slowly varying episodes act as state dependence. Only the models with the space-time field bring the estimate close to zero, with coverage 0.90: $\hat{\delta}$ is 0.05 with four-week steps and 0.01 with weekly steps. The mixed scenario behaves in the same way. The static field leaves an upward bias (0.78 against 0.5), which the space-time field removes (0.55 and 0.54, coverage 0.85 and 0.80).

Resolving v at four-week steps instead of weekly steps costs little bias in these scenarios, and it is four to five times faster. The space-time field has an AR(1) coefficient of 0.9 per week, so it changes slowly relative to the four-week history window. Finally, the XGBoost importance of the lag feature is about the same under all three truths (share 0.09–0.12). The variable importance of recent history therefore says nothing about whether violence begets violence.

4.6. *S4: biome-specific climate effects.* The temperature effect follows (3.4), with biome means $\boldsymbol{\gamma} = (-0.3, -0.1, 0.1, 0.3)$ and a within-biome Matérn deviation η ($\sigma_\eta = 0.15$, $\rho_\eta = 600$ km), $\sigma_w = 0.8$ and 30 replicates. We compare:

- fixed biome-specific slopes, without and with $u + w$;
- the SVC model (3.4);
- two XGBoost summaries fitted to the anomaly: the per-biome least-squares slope of the SHAP value of the anomaly on the anomaly, and the same slope computed from the SHAP value plus the interaction values credited to the anomaly ($\Phi_{jj} + 2\sum_{k \neq j} \Phi_{jk}$ in the notation of Lundberg et al., 2020). For a pure biome-by-temperature interaction, the first targets $\gamma_b/2$ and the second γ_b;

TABLE 9

S4: estimation of biome-specific temperature effects (30 replicates). "Detects sign": proportion of replicates in which the interval excludes zero on the correct side. ‡Median SHAP value of raw temperature within the biome, with the interquartile range of the bootstrap replicates, as in the original analysis; it estimates neither γ_b nor $\bar{\beta}_b$.

Estimator / biome	Mean est.	Cov. γ_b	Cov. $\bar{\beta}_b$	Detects sign
Biome slopes, no latent effects				
biome 1 ($\gamma = -0.3$)	-0.273	0.57	0.77	1.00
biome 2 ($\gamma = -0.1$)	-0.083	0.57	0.83	0.60
biome 3 ($\gamma = +0.1$)	0.108	0.53	0.70	0.73
biome 4 ($\gamma = +0.3$)	0.307	0.50	0.53	1.00
Biome slopes + u + w				
biome 1 ($\gamma = -0.3$)	-0.286	0.53	0.87	1.00
biome 2 ($\gamma = -0.1$)	-0.082	0.57	0.90	0.67
biome 3 ($\gamma = +0.1$)	0.119	0.60	0.73	0.73
biome 4 ($\gamma = +0.3$)	0.319	0.53	0.87	1.00
SVC: biome means + η + u + w				
biome 1 ($\gamma = -0.3$)	-0.277	0.97	1.00	0.93
biome 2 ($\gamma = -0.1$)	-0.077	0.93	1.00	0.23
biome 3 ($\gamma = +0.1$)	0.111	0.90	1.00	0.33
biome 4 ($\gamma = +0.3$)	0.309	0.83	1.00	0.97
XGBoost SHAP slope (targets $\gamma_b/2$)				
biome 1 ($\gamma = -0.3$)	-0.092	0.03	0.03	0.60
biome 2 ($\gamma = -0.1$)	-0.026	0.47	0.53	0.23
biome 3 ($\gamma = +0.1$)	0.048	0.47	0.40	0.37
biome 4 ($\gamma = +0.3$)	0.091	0.03	0.00	0.60
XGBoost SHAP slope, interactions credited				
biome 1 ($\gamma = -0.3$)	-0.191	0.27	0.40	0.83
biome 2 ($\gamma = -0.1$)	-0.056	0.63	0.63	0.30
biome 3 ($\gamma = +0.1$)	0.089	0.57	0.73	0.53
biome 4 ($\gamma = +0.3$)	0.176	0.40	0.30	0.80
XGBoost median SHAP, raw temperature‡				
biome 1 ($\gamma = -0.3$)	-0.052	0.03	0.07	0.57
biome 2 ($\gamma = -0.1$)	-0.151	0.13	0.13	0.97
biome 3 ($\gamma = +0.1$)	-0.150	0.00	0.00	0.00
biome 4 ($\gamma = +0.3$)	-0.043	0.00	0.00	0.20

- the statistic of the original analysis: the median SHAP value of raw temperature within each biome, from a model fitted to raw temperature.

The XGBoost models are original-like. Their intervals come from 20 replicates of the 2.5° block bootstrap on a fixed 30% subsample; for the median SHAP value we report the interquartile range of the bootstrap replicates, the criterion of the original analysis. Coverage is reported for both the biome mean γ_b and the realised biome average $\bar{\beta}_b$ (Section 3.5).

The SVC model is the only estimator with close to nominal coverage for the biome means γ_b (0.83–0.97; Table 9). Its intervals are conservative for the realised biome averages $\bar{\beta}_b$ (coverage 1.00), as expected, because they also account for the sampling of regions within biomes. Fixed biome slopes are nearly unbiased, but their intervals ignore heterogeneity within biomes. They cover γ_b in only 50–60% of replicates, and $\bar{\beta}_b$ in 53–90%. The price of honest intervals is power: the SVC model detects the small effects ($|\gamma_b| = 0.1$) in only 23–33% of replicates. The SVC model also recovers the coefficient surface (Table 10). The correlation between $\hat{\beta}(s)$ and $\beta(s)$ is 0.92, the RMSE is 28% lower than with biome slopes only (0.106 against 0.147), and $\hat{\sigma}_\eta = 0.13$ against a true 0.15.

TABLE 10
S4: recovery of the spatially varying coefficient $\beta(s)$ (mean and SD over 30 replicates).

Quantity	Mean	SD
$\mathrm{cor}\{\hat{\beta}(s), \beta(s)\}$, SVC	0.918	0.036
RMSE of $\hat{\beta}(s)$, SVC	0.106	0.010
RMSE of $\hat{\beta}(s)$, biome slopes only	0.147	0.020
$\hat{\sigma}_\eta$ [truth 0.15]	0.133	0.025
$\hat{\rho}_\eta$ (km) [truth 600]	563.704	261.148

The XGBoost summaries are much less informative:

- *SHAP slopes.* The per-biome slopes of the ordinary SHAP values have the right ordering but are about $0.3\gamma_b$, below even the $\gamma_b/2$ expected from the splitting of interactions. Crediting the interaction values to temperature raises them to about $0.6\gamma_b$. The remaining attenuation reflects the shrinkage built into boosting and the early stopping of the original procedure.
- *Median SHAP values.* The median SHAP value of raw temperature within a biome, the statistic of the original analysis, is negative in *every* biome (-0.04 to -0.15), whatever the true effect. Its bootstrap interquartile range excludes zero with a negative sign in every replicate in biome 3 and in 53% of replicates in biome 4, where the true effects are positive. Applying the original criterion would therefore report a "negative contribution of temperature" where the effect is positive.

As argued in Section 3.5, the median SHAP value describes where a biome's covariate values lie relative to the model baseline, not the strength of the effect.

5. Application to ACLED 2018–2024. All models are trained on 2018–2022 and evaluated on 2023–2024. The boosting models use the features and hyperparameters of Menicali, Marcantonio and Castruccio (2026). The computations were run at the Notre Dame Center for Research Computing.

5.1. *Reconstructing the cell-week table.* The table built by the original pipeline has 70,033,444 rows but only 34,671,180 distinct cell-weeks (Section 2.1). In one partition, 613,459 cell-weeks appear once, while others appear 2, 3, 8, 16, 32, 48 and up to 648 times. The national indicators and IPC regions are attached by spatial intersection and then merged on cell and year. A cell that touches several countries or IPC regions is therefore multiplied at every join, and each copy combines indicators from different units.

The duplication matters for more than the covariates. The four-week event counts of the original pipeline are computed row by row within each cell, after sorting by cell and week. When a cell-week has copies, the rows immediately before a given row are copies of the same week, so the "past" count contains the current week's events, which are the outcome. We recomputed the original counts exactly and compared them with the true counts over the four previous weeks. In 22% of event weeks, for both outcomes, the original count includes the current week's events while the true past count is zero. Used directly as a predictor of the outcome, without any model, the original count reaches an AUC-PR of 0.883 for conflict and 0.833 for unrest. The true count reaches 0.495 and 0.450. The leak is concentrated in the duplicated cells, which lie mostly along borders and coasts.

The repaired table has one row per cell-week (Section 2.1). After the biome exclusion it has 59,344 cells and 366 weeks. ACLED coverage removes [TODO: x] further cell-weeks. The test period has [TODO: x] cell-weeks in [TODO: x] cells with complete climate records.

TABLE 11

XGBoost on the 2023–2024 test period: as published (duplicated table, early stopping on the test years), corrected (one row per cell-week, ACLED coverage, early stopping on 2022) and corrected with a cell event-rate feature. AUC-PR is also reported separately for cells that were duplicated in the original table and for the others.

Pipeline	Outcome	Test rows	Prevalence	AUC-PR	Border cells	Interior cells	Persistence
As published	Conflict	14.43M	2.98%	0.934	0.966	0.498	0.890
As published	Unrest	14.43M	3.58%	0.917	0.968	0.528	0.852
Corrected	Conflict	4.79M	1.62%	0.549	0.588	0.528	0.511
Corrected	Unrest	4.79M	2.73%	0.553	0.542	0.555	0.500

5.2. *The boosting analysis before and after the repair.* Table 11 reproduces the XGBoost analysis on the duplicated table and then repeats it on the repaired table. In the corrected version, the number of trees is chosen by early stopping on 2022 after training on 2018–2021, rather than on the test years, and the class weight is removed. On the duplicated table we recover the published performance, with AUC-PR [TODO: 0.934] for conflict and [TODO: 0.917] for unrest, against 0.93 and 0.91 reported. Split by cell type, nearly all of it comes from the duplicated cells (AUC-PR [TODO: 0.97]), and on the other cells AUC-PR is [TODO: 0.50–0.53]. On the repaired table AUC-PR falls to [TODO: x] for conflict and [TODO: x] for unrest. The persistence baseline reaches [TODO: x] and [TODO: x]. The location-aware boosting model, which adds each cell's training event rate, reaches [TODO: x] and [TODO: x]. [TODO: Interpretation: how much the covariates add to recent history, and how much a cell intercept adds to the covariates.]

5.3. *Residual spatial structure and the hybrid model.* We fit the hybrid model (3.5) to Africa and the Middle East (longitude 20°W to 65°E, latitude 36°S to 45°N; [TODO: x] cells). It combines the XGBoost offset cross-fitted by country, country effects, and either a Matérn field with sea barriers (Remark 1) or independent cell effects. Table 12 reports the test-period residual diagnostics of every model, and Figure 4 the semivariograms of the cross-fitted offset's residuals.

[TODO: Residual structure of the cross-fitted offset (Moran's I, within-country and across-border correlations, semivariogram) for conflict and unrest; and how much of it remains after the hybrid models.]

[TODO: Fitted hyperparameters: range and standard deviation of the fields; whether the range is resolved by the 0.5° grid (Section 3.2).]

The estimated fields are shown in Figure 3, and Table 13 summarises the posterior of the field averaged over regions. [TODO: Regions with posterior probability of a positive regional average close to one; the clearest negative regions. Interpretation as persistent *recorded* risk, mixing occurrence and reporting (Section 2.1).]

Table 14 compares out-of-sample scores on identical test rows, with log-score differences against the corrected XGBoost model and their cell-clustered standard errors. [TODO: Results: hybrid_field against hybrid_iid (does the spatial smoothing matter beyond a cell intercept?), against the location-aware boosting baseline, and against persistence. Global fit.]

5.4. *State dependence, climate and national indicators.* [TODO: Results of the parametric models: (i) the state-dependence coefficient δ with lags only, with $u + w$, and with the space-time field v in the central Sahel, read as conditional dependence of recorded events (Section 3.4); (ii) biome means and within-biome variation of the maximum-temperature effect from the SVC model, against fixed biome slopes; (iii) the between- and within-country coefficients of the national indicators without latent effects, with u, with $u + w$, and with spatial+ for $m = 10$, 20 and m^*, read in the light of Section 4.4.]

TABLE 12

Test-period residual diagnostics, Africa and the Middle East: Moran's I of the standardised cell residuals (permutation p-value, bounded below by 0.001) and the residual correlation between neighbouring cells in the same country and across a border; and the hyperparameters of the hybrid model.

Outcome	Moran's I	Neighbour residual corr.		$\hat{\rho}_w$ (km)	$\hat{\sigma}_w$	$\hat{\sigma}_u$
		within country	across border			
Conflict	0.452	0.482	0.305	277	1.24	0.35
Unrest	0.110	0.114	0.108	72	1.11	0.59

TABLE 13

Posterior summaries of the field of the hybrid model averaged over regions: mean, standard deviation and posterior probability that the regional average is positive.

[TODO: table from summary.json (boxes)]

TABLE 14

Out-of-sample scores, Africa and the Middle East, 2023–2024, on identical test rows. Lower is better for the log and Brier scores. Differences in mean log score against the corrected XGBoost model, with standard errors clustered by cell.

Outcome	Model	Log score	Brier	AUC-PR	AUC-ROC
Conflict	Cross-fitted XGBoost	0.0630	0.01608	0.508	0.942
Conflict	Cross-fitted XGBoost + $u + w$	0.0600	0.01595	0.527	0.958
Conflict	Persistence	–	–	0.228	–
Unrest	Cross-fitted XGBoost	0.0672	0.01747	0.522	0.946
Unrest	Cross-fitted XGBoost + $u + w$	0.0633	0.01698	0.548	0.959
Unrest	Persistence	–	–	0.233	–

6. Discussion.

[TODO: Discussion (not drafted yet).]

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SUPPLEMENTARY MATERIAL

Code

Python implementation of the latent Gaussian model, the simulator and all simulation studies.

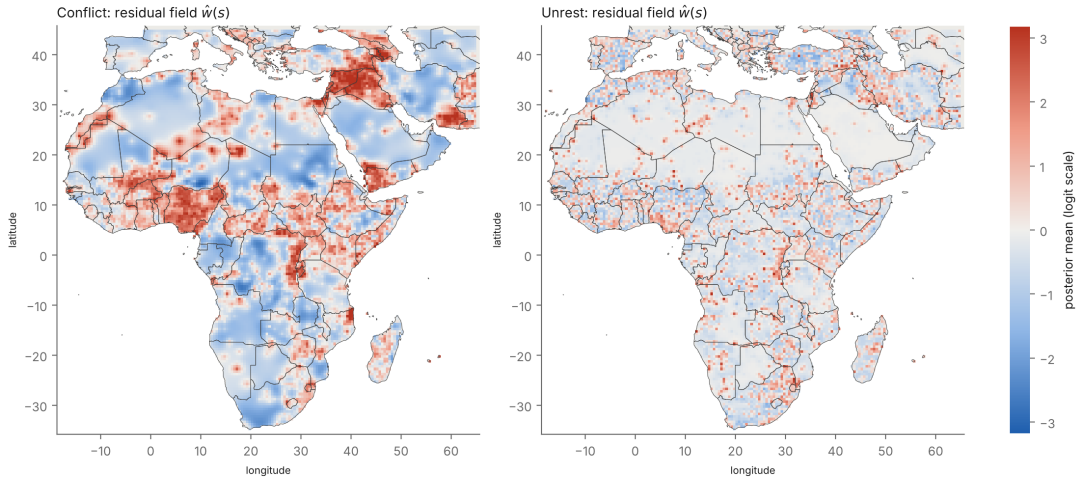


FIG 3. *Posterior mean of the residual field $w(s)$ of the hybrid model for conflict (left) and unrest (right), on the logit scale. Positive values mark persistently higher recorded risk than predicted by the covariates, recent event history and country effects.*

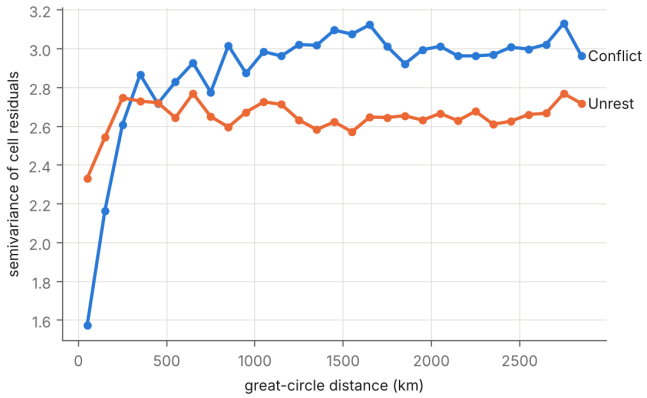


FIG 4. *Empirical semivariograms of the standardised test-period cell residuals of the XGBoost model cross-fitted by country, against great-circle distance.*

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